

Unit 1

Unit 1: HTML, CSS, Bootstrap & W3C — Practical/Oral Exam Notes

Context

Part of [Web Application Development](#) | Practical + Oral Exam Prep | SPPU Third Year IT

High-Yield Oral Topics

- **HTML5 Semantic Elements** — Examiners love this. Know 6+ examples.
- **CSS Flexbox vs Grid** — Always compared in viva.
- **Bootstrap Grid (col-md-6)** — Must explain responsiveness.
- **CSS Box Model** — Margin, Border, Padding, Content.
- **Inline vs Block vs Inline-Block** — Trap question.

MINI PROJECT ANCHOR: Student Management System (SMS)

All examples in this unit use the **Student Management System** for consistency.

- Frontend: HTML + CSS + Bootstrap
- Features: Student Registration Form, Dashboard, Student List Table

SECTION 1 — HTML

Exam Definition

HTML (HyperText Markup Language) is the **standard markup language** used to create the structure and content of web pages. It uses **tags** to define elements like headings, paragraphs, forms, and images.

Core Concept

HTML is NOT a programming language — it is a **markup language**. It tells the browser *what* to display, not *how* to behave. CSS handles appearance; JavaScript handles behavior.

Must-Know Tags (Viva Favorite)

Category	Tags	Purpose
Structure	<code><html></code> , <code><head></code> , <code><body></code>	Page skeleton

Category	Tags	Purpose
Headings	<h1> to <h6>	Hierarchy of text
Text	<p> , , , 	Content
Links		Navigation
Lists	, ,	Ordered/Unordered
Tables	<table> , <tr> , <th> , <td>	Tabular data
Forms	<form> , <input> , <button> , <select>	User input
Media	 , <video> , <audio>	Rich media
Semantic	<header> , <nav> , <section> , <article> , <footer>	Meaningful structure

 Practical Code — SMS Registration Form

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Registration - SMS</title>
</head>
<body>
  <header>
    <h1>Student Management System</h1>
    <nav>
      <a href="index.html">Home</a> |
      <a href="students.html">Students</a>
    </nav>
  </header>

  <main>
    <section id="registration">
      <h2>Register New Student</h2>
      <form id="studentForm" action="/register" method="POST">
        <label for="name">Full Name:</label>
        <input type="text" id="name" name="name" required placeholder="Enter full name">

        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>

        <label for="course">Course:</label>
        <select id="course" name="course">
          <option value="IT">Information Technology</option>
```

```
<option value="CS">Computer Science</option>
</select>

<label for="dob">Date of Birth:</label>
<input type="date" id="dob" name="dob">

<button type="submit">Register</button>
<button type="reset">Clear</button>
</form>
</section>

<section id="studentTable">
  <h2>Registered Students</h2>
  <table border="1">
    <thead>
      <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Course</th>
        <th>Email</th>
      </tr>
    </thead>
    <tbody id="tableBody">
      <!-- Populated by JavaScript -->
    </tbody>
  </table>
</section>
</main>

<footer>
  <p>&copy; 2025 SMS – Student Management System</p>
</footer>
</body>
</html>
```

🔥 Semantic HTML5 — MOST ASKED IN VIVA

Semantic Tag	Purpose	Non-Semantic Equivalent
<header>	Top section of page/section	<div id="header">
<nav>	Navigation links	<div id="nav">
<main>	Main content of page	<div id="main">
<section>	Thematic grouping of content	<div>

Semantic Tag	Purpose	Non-Semantic Equivalent
<article>	Self-contained content (blog post)	<div>
<aside>	Sidebar / related content	<div id="sidebar">
<footer>	Bottom of page	<div id="footer">
<figure>	Image with caption	<div>

Why Semantic HTML? — SEO (Search Engine Optimization), Accessibility (screen readers), Code readability.

HTML Attributes — Viva Quick Fire

Attribute	Used On	Purpose
href	<a>	Link destination
src	 , <script>	File source
alt		Alternate text (accessibility)
action	<form>	Where to send form data
method	<form>	GET or POST
type	<input>	text, email, password, date, etc.
required	<input>	HTML5 validation
id	Any	Unique identifier
class	Any	CSS styling hook

VIVA Q&A — HTML

Q1: What is the difference between <div> and ?

<div> is a **block-level** element — takes full width, starts on new line. Used for layout sections. is an **inline** element — only as wide as its content. Used to style a portion of text.

Q2: What is the difference between GET and POST in forms?

GET appends data to URL (visible, used for search). **POST** sends data in request body (hidden, secure, used for login/registration). POST is preferred for sensitive data like passwords.

Q3: What are HTML5 Semantic elements and why use them?

HTML5 semantic elements like <header> , <nav> , <article> give **meaning** to the structure. Benefits: Better SEO (Google understands page structure), Accessibility (screen readers), and cleaner code.

Q4: What is the DOCTYPE declaration?

<!DOCTYPE html> tells the browser to render the page in **HTML5 standards mode**. Without it, browsers go into "quirks mode" which causes inconsistent rendering.

Q5: What is alt attribute in ?

The alt attribute provides **alternative text** that displays if the image fails to load. It is also used by **screen readers** for visually impaired users and improves **SEO**.

⚠️ Common Mistakes Students Make

- Forgetting <!DOCTYPE html> at top
- Not closing tags (especially , <td>)
- Using instead of (semantic vs presentational)
- Not adding name attribute to form inputs (data won't be sent)
- Mixing up id (unique) vs class (reusable)

💡 Memory Trick

"Semantic HTML = You Know What's Inside Without Opening the Box"

<article> → article content. <nav> → navigation. <footer> → bottom info.

SECTION 2 — CSS

📌 Exam Definition

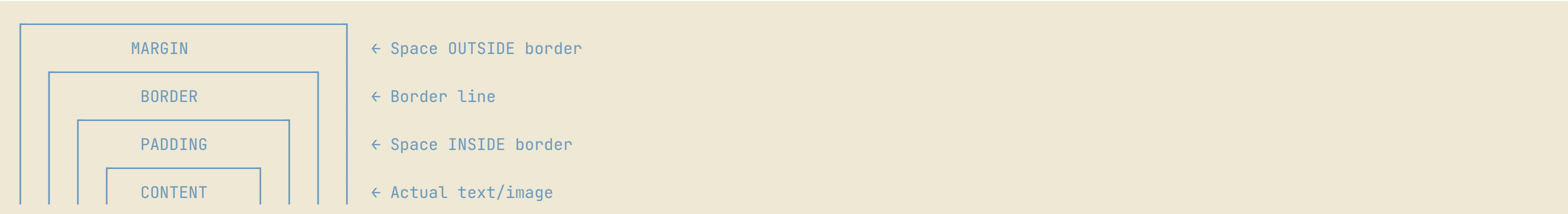
CSS (Cascading Style Sheets) is a **stylesheet language** used to control the presentation (layout, colors, fonts, spacing) of HTML elements. It **separates design from structure**.

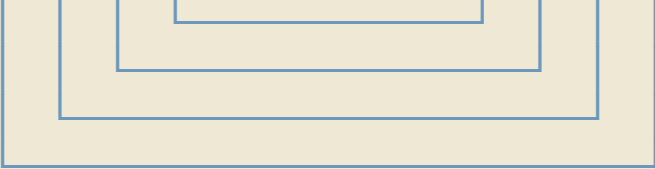
🔑 Three Types of CSS

Type	Syntax	Priority	Use Case
Inline	style="color:red" inside tag	Highest	Quick one-off styles
Internal	<style> inside <head>	Medium	Single-page styles
External	<link rel="stylesheet" href="style.css">	Normal	Best practice, reusable

Cascade Rule: Inline > Internal > External (when specificity is equal)

📦 The CSS Box Model — MUST KNOW





Viva Answer: "Margin is OUTSIDE the border, Padding is INSIDE the border. Margin creates space between elements; padding creates space between content and its border."

Practical Code — SMS Dashboard Styling

```
/* External CSS: style.css */

/* Reset defaults */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

/* Body styling */
body {
  font-family: 'Arial', sans-serif;
  background-color: #f0f4f8;
  color: #333;
}

/* Header */
header {
  background-color: #2c3e50;
  color: white;
  padding: 15px 20px;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

/* Navigation */
nav a {
  color: #ecf0f1;
  text-decoration: none;
  margin-left: 20px;
  font-weight: bold;
}
```



```
nav a:hover {
  color: #3498db;
  transition: color 0.3s ease;
}

/* Dashboard Cards - FLEXBOX */
.card-container {
  display: flex;
  flex-wrap: wrap;
  gap: 20px;
  padding: 20px;
}

.card {
  background: white;
  border-radius: 8px;
  padding: 20px;
  flex: 1;
  min-width: 200px;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  border-left: 4px solid #3498db;
  transition: transform 0.2s ease;
}

.card:hover {
  transform: translateY(-3px);
}

/* Grid Layout for main content */
.main-grid {
  display: grid;
  grid-template-columns: 250px 1fr;
  gap: 20px;
  padding: 20px;
}

/* Sidebar */
.sidebar {
  background: white;
  border-radius: 8px;
  padding: 15px;
}

/* Form Styling */
```

```
form input, form select {
  width: 100%;
  padding: 10px;
  margin-bottom: 15px;
  border: 1px solid #ddd;
  border-radius: 5px;
  font-size: 14px;
}

form button {
  background-color: #3498db;
  color: white;
  padding: 10px 25px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
  font-size: 16px;
  transition: background 0.3s ease;
}

form button:hover {
  background-color: #2980b9;
}

/* Animation Example */
@keyframes fadeIn {
  from { opacity: 0; transform: translateY(20px); }
  to   { opacity: 1; transform: translateY(0); }
}

.card {
  animation: fadeIn 0.5s ease forwards;
}

/* Media Query – Responsive */
@media (max-width: 768px) {
  .main-grid {
    grid-template-columns: 1fr;
  }
  .card-container {
    flex-direction: column;
  }
}
```


Flexbox vs Grid — MOST COMPARED IN VIVA

Feature	Flexbox	Grid
Dimension	1-Dimensional (row OR column)	2-Dimensional (rows AND columns)
Use Case	Navigation bars, card rows, centering	Page layout, dashboards
Direction	<code>flex-direction: row/column</code>	<code>grid-template-columns/rows</code>
Alignment	<code>justify-content</code> , <code>align-items</code>	<code>justify-items</code> , <code>align-items</code>
Example	Navigation links in a row	3-column dashboard

CSS Positioning — Viva Trap

Value	Behavior
<code>static</code>	Default. Normal flow. <code>top/left</code> ignored.
<code>relative</code>	Offset from its NORMAL position. Other elements not affected.
<code>absolute</code>	Removed from flow. Positioned relative to nearest positioned parent.
<code>fixed</code>	Removed from flow. Fixed to browser viewport (sticky navbar).
<code>sticky</code>	Normal flow until scroll threshold, then fixed.

VIVA Q&A — CSS

Q1: What is specificity in CSS?

Specificity determines which CSS rule applies when multiple rules target the same element. Priority: Inline style (1000) > ID selector (100) > Class selector (10) > Element selector (1). Higher specificity wins.

Q2: What is the difference between `margin` and `padding`?

Margin is the space **outside** the border (between elements). Padding is the space **inside** the border (between content and border). Background color fills the padding area but NOT the margin.

Q3: What is `box-sizing: border-box`?

By default, width/height only includes content. With `border-box`, width/height includes padding and border too. This makes layout calculation much easier and predictable.

Q4: How does CSS animation work?

CSS animation uses `@keyframes` to define style changes at different points, and `animation` property applies it to an element. Example: `animation: fadeIn 0.5s ease` — the element fades in over 0.5 seconds.

Memory Tricks

- **Box Model:** "My Big Precious Content" = Margin, Border, Padding, Content (outside to inside)
- **Flex is 1D, Grid is 2D** → Flex = one direction at a time, Grid = full table layout

SECTION 3 — BOOTSTRAP

Exam Definition

Bootstrap is a **free, open-source CSS framework** that provides pre-built responsive components and a **12-column grid system**, allowing developers to build mobile-responsive web pages rapidly without writing CSS from scratch.

Why Bootstrap over Raw CSS?

Aspect	Raw CSS	Bootstrap
Time	Write everything from scratch	Pre-built components ready
Responsiveness	Manual media queries	Built-in grid system
Browser Compatibility	Test/fix manually	Handled by Bootstrap
Consistency	Depends on developer	Consistent across team
Learning Curve	Deep CSS knowledge needed	Learn classes, apply fast

Bootstrap Grid System — MUST KNOW

Bootstrap divides every row into **12 columns**. You specify how many columns an element takes at each breakpoint.

```
<!-- Setup: Add Bootstrap via CDN -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Grid Example: SMS Dashboard Layout -->
<div class="container">
  <div class="row">
    <!-- Sidebar: 3 columns on desktop, full width on mobile -->
    <div class="col-12 col-md-3">
      <div class="card">
        <div class="card-body">
          <h5>Navigation</h5>
          <!-- Sidebar content -->
        </div>
      </div>
    </div>
    <!-- Main Content: 9 columns on desktop -->
    <div class="col-12 col-md-9">
      <!-- Statistics Cards -->
      <div class="row mb-4">
        <div class="col-md-4">
```

```
<div class="card text-white bg-primary">
  <div class="card-body">
    <h5 class="card-title">Total Students</h5>
    <h2>150</h2>
  </div>
</div>
<div class="col-md-4">
  <div class="card text-white bg-success">
    <div class="card-body">
      <h5 class="card-title">Active Courses</h5>
      <h2>12</h2>
    </div>
  </div>
</div>
<div class="col-md-4">
  <div class="card text-white bg-warning">
    <div class="card-body">
      <h5 class="card-title">Pending Fees</h5>
      <h2>23</h2>
    </div>
  </div>
</div>
</div>

<!-- Student Table -->
<table class="table table-striped table-hover table-bordered">
  <thead class="table-dark">
    <tr>
      <th>Roll No</th><th>Name</th><th>Course</th><th>Action</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>101</td><td>Rahul Sharma</td><td>IT</td>
      <td>
        <button class="btn btn-sm btn-primary">Edit</button>
        <button class="btn btn-sm btn-danger">Delete</button>
      </td>
    </tr>
  </tbody>
</table>
</div>
```

```
</div>
</div>
```

Bootstrap Breakpoints

Breakpoint	Code	Width	Target
Extra Small	col-	< 576px	Phones
Small	col-sm-	≥ 576px	Large phones
Medium	col-md-	≥ 768px	Tablets
Large	col-lg-	≥ 992px	Laptops
Extra Large	col-xl-	≥ 1200px	Desktops

Viva Tip: col-md-6 means "take 6 of 12 columns on medium screens and above" = **two equal columns** on tablet/desktop.

Key Bootstrap Classes — Cheatsheet

Class	Purpose
container / container-fluid	Centered wrapper / Full-width wrapper
row	Flex row (parent of columns)
col-md-6	Column taking 6/12 width on medium+
btn btn-primary	Blue button
table table-striped	Striped table
card, card-body, card-title	Card component
navbar navbar-expand-lg	Responsive navigation bar
d-flex justify-content-center	Flexbox centering
mt-3, mb-4, px-2	Margin/Padding utilities
text-center, fw-bold	Text alignment/weight
d-none d-md-block	Hide on mobile, show on desktop

VIVA Q&A — Bootstrap

Q1: What is the Bootstrap grid system?

Bootstrap's grid system uses a series of containers, rows, and columns to layout content. It's based on a **12-column** flexible grid. You specify how many columns an element spans using classes like col-md-6 (6 of 12 = half width on medium+ screens).

Q2: How does Bootstrap achieve responsiveness?

Bootstrap uses **CSS media queries** internally. When you use col-md-6 col-12, Bootstrap applies display: block; width: 100% on small screens and width: 50% on medium+ screens automatically.

Q3: What is the difference between `container` and `container-fluid`?

`container` has a fixed max-width that changes at breakpoints (centered with margins). `container-fluid` is always 100% wide — it spans the full viewport width.

Q4: Why use Bootstrap instead of writing CSS from scratch?

Bootstrap saves development time with pre-built components (buttons, navbars, modals, tables), ensures cross-browser compatibility, and provides a proven responsive grid system. It's ideal for rapid prototyping and team projects.

 Oral Trap Questions

Trap: "Can you use Bootstrap AND custom CSS together?"

Yes! You add Bootstrap via CDN, then write your own CSS file after it. Your custom styles override Bootstrap where needed. This is the standard approach.

Trap: "Bootstrap uses JavaScript — what for?"

Bootstrap includes JavaScript (via Popper.js) for **interactive components** like modals, dropdowns, carousels, tooltips, and collapsible navbars. Pure CSS cannot handle these interactions.

SECTION 4 — W3C

 Exam Definition

W3C (World Wide Web Consortium) is the **international standards organization** that develops and maintains open web standards to ensure long-term growth and interoperability of the web.

Key Points for Viva

Aspect	Details
Founded by	Tim Berners-Lee (inventor of WWW) in 1994
Purpose	Set standards for HTML, CSS, XML, Web Accessibility
Key Standards	HTML5, CSS3, WCAG (Accessibility), SVG, WebRTC
Why Important	Ensures all browsers render the same code consistently

 VIVA Q&A — W3C

Q1: What is W3C and why is it important?

W3C (World Wide Web Consortium) is the body that creates and maintains web standards. Without W3C, each browser would implement HTML and CSS differently, breaking cross-browser compatibility. W3C ensures that code written by one developer works consistently everywhere.

Q2: What is WCAG?

WCAG (Web Content Accessibility Guidelines) is a W3C standard that defines how to make web content accessible to people with disabilities (visual, auditory, cognitive). It defines criteria like sufficient color contrast and alt text for images.

PRACTICAL FILE EXPECTATIONS — Assignment 1

Your practical file for Assignment 1 should show:

- 1. **HTML Registration Form** — Student registration with name, email, course, DOB fields
- 2. **Bootstrap Dashboard** — Sidebar + 3 statistics cards + student table
- 3. **CSS Styling** — External stylesheet with responsive design
- 4. **JavaScript Validation** — Form validation + store data in `localStorage`
- 5. **AJAX POST** — Submit form data via AJAX and display success message
- 6. **Student List Page** — Read from localStorage and display in table

Viva Flow for Assignment 1:

"I created a Student Management System frontend. The registration page uses **Bootstrap grid** (col-md-3 for sidebar, col-md-9 for content). The form uses **HTML5 validation** (required, type="email"). On submit, **JavaScript** validates the data and stores it in **localStorage** as a JSON array. I also used **AJAX POST** to send the data to the backend. The dashboard shows student statistics using **Bootstrap cards**."

High Probability Exam Questions

- 1. What is the difference between HTML and HTML5? (Answer: HTML5 added semantic elements, audio/video support, canvas, local storage, form validation)
- 2. Explain the CSS Box Model with diagram
- 3. What is Flexbox? How is it different from Grid?
- 4. What is Bootstrap Grid? Explain with `col-md-6` example
- 5. What are semantic elements? Give 5 examples
- 6. Difference between `margin` and `padding`
- 7. What is media query? How does it work?
- 8. What is specificity in CSS?
- 9. What is `DOCTYPE` ?
- 10. What is the difference between `id` and `class` in CSS?

Quick Revision Table

Topic	Key Point	Viva One-Liner
HTML	Markup language for structure	"HTML tells browser WHAT to show"

Topic	Key Point	Viva One-Liner
CSS	Stylesheet for presentation	"CSS tells browser HOW to show it"
Bootstrap	CSS framework with 12-col grid	"Responsive design without writing media queries"
Semantic HTML	Tags with meaning	"Screen readers and SEO benefit"
Flexbox	1D layout	"Row OR column alignment"
Grid	2D layout	"Rows AND columns layout"
Box Model	M-B-P-C	"Margin outside , Padding inside border"
W3C	Standards body	"Tim Berners-Lee's web standards org"